Gupta's Sandwich

Restaurant Management System

Complete Documentation

Technical Documentation Team

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Contents

1 System Overview

Gupta's Sandwich is a comprehensive restaurant management system designed to handle multi-outlet operations. The system manages:

- **Multiple Outlets**: Support for various restaurant locations (e.g., Koregaon Park, Baner, Kothrud)
- **Menu Management**: Centralized menu with outlet-specific availability
- **Order Processing**: Dine-in, parcel, and online orders (Swiggy/Zomato)
- **Kitchen Workflow**: KOT-based kitchen order management
- **Customer Management**: Track customer history and preferences
- **Financial Tracking**: Accounting entries and payment management
- **Reporting**: Dynamic sales, financial, and operational reports

2 Database Schema

2.1 ADMIN Table

Administrator table for system-wide management.

Field	Type	Description
id	INT	Primary key
name	VARCHAR	Admin name

Example Data:

id	name
1	Super Admin

2.2 OUTLET Table

Stores information about each restaurant location.

Field	Type	Description
id	INT	Primary key
name	VARCHAR	Outlet name (e.g., "Koregaon Park")
address	TEXT	Complete address of the outlet
manager	VARCHAR	Outlet manager name
phone	VARCHAR	Contact phone number
email	VARCHAR	Outlet email address
username	VARCHAR	Login username for outlet
password	VARCHAR	Login password (store hashed)
status	ENUM	'active' or 'inactive'

Example Data:

id	name	address	manager	phone	
1	Koregaon Park	Shop 12, Lane 5, Koregaon Park, Pune	Ramesh Gupta	9876543210	kp©
2	Baner	Plot 8, Baner Road, Baner, Pune	Suresh Sharma	9876543211	baner
3	Kothrud	Near Vanaz, Kothrud, Pune	Dinesh Patil	9876543212	kothru

2.3 USER Table (Outlet Staff)

Manages staff members working at each outlet.

Field	Type	Description
id	INT	Primary key
outlet_id	INT	Foreign key to OUTLET
name	VARCHAR	Staff member name
email	VARCHAR	Email address
username	VARCHAR	Login username
password	VARCHAR	Login password (store hashed)
roleLabel	VARCHAR	Display role name (e.g., "Cashier")
appRole	VARCHAR	System role (e.g., "Staff", "Manager")
permissions	JSON	Object defining access permissions
status	ENUM	'active' or 'inactive'

Permissions Structure:

```
{
    "pos": true,
    "reports": true,
    "inventory": false,
    "staff_management": false
}
```

Example Data:

id	outlet_id	name	email	roleLabel	appRole	status
101	1	Ankit Sharma	ankit@guptasandwich.in	Cashier	Staff	active

2.4 MENU_ITEM Table (Dishes)

Central repository of all dishes offered across outlets.

Field	Type	Description
id	INT	Primary key
name	VARCHAR	Dish name
category	VARCHAR	Category (e.g., "Sandwiches", "Subs", "Beverages")
dine_in_price	DECIMAL	Price for dine-in customers
parcel_price	DECIMAL	Price for takeaway/parcel
swiggy_price	DECIMAL	Price on Swiggy platform (NULL if unavailable)
zomato_price	DECIMAL	Price on Zomato platform (NULL if unavailable)
ingredients	TEXT	Description of ingredients
outlets	VARCHAR	Outlet availability summary
available_swiggy	BOOLEAN	Whether available on Swiggy
available_zomato	BOOLEAN	Whether available on Zomato
is_active	BOOLEAN	Whether dish is active in system

Example Data:

id	name	category	dine_in	parcel	swiggy	zomato	is_active
1	Veg Club Sandwich	Sandwiches	120	110	140	140	true
2	Chicken Tikka Sub	Subs	180	170	210	210	true
3	Paneer Griller	Sandwiches	150	140	175	175	true
4	Cold Coffee	Beverages	80	80	95	95	true
5	Egg Bhurji Sandwich	Sandwiches	130	120	150	150	true
6	Masala Chai	Beverages	40	40	NULL	NULL	true

2.5 MENU_ITEM_OUTLET Table (Junction)

Controls which dishes are available at which outlets.

Field	Type	Description
id	INT	Primary key
menu_item_id	INT	Foreign key to MENU_ITEM
outlet_id	INT	Foreign key to OUTLET
is_available	BOOLEAN	Whether dish is available at this outlet

Example Data:

id	menu_item_id	outlet_id	is_available
1	1	1	true
2	1	2	true
3	1	3	true
4	2	1	true
5	2	2	true
6	3	1	true

2.6 INGREDIENT Table

Inventory ingredients used in dishes.

Field	Type	Description
id	INT	Primary key
name	VARCHAR	Ingredient name
unit	VARCHAR	Unit of measurement (pcs, g, ml)

Example Data:

id	name	unit
1	Bread	pcs
2	Lettuce	g
3	Tomato	g
4	Mayo	g
5	Sub Roll	pcs
6	Chicken	g
7	Onion	g

2.7 MENU_ITEM_INGREDIENT Table (Junction)

Recipe mapping - which ingredients and how much go into each dish.

Field	**Type**	**Description**
id	INT	Primary key
menu_item_id	INT	Foreign key to MENU_ITEM
ingredient_id	INT	Foreign key to INGREDIENT
quantity	DECIMAL	Quantity required

2.8 TABLES Table

Physical tables in dining areas for dine-in orders.

Field	**Type**	**Description**
id	INT	Primary key
outlet_id	INT	Foreign key to OUTLET
table_number	VARCHAR	Table identifier (e.g., "T1", "T2")
capacity	INT	Seating capacity
location	VARCHAR	Area description
is_active	BOOLEAN	Whether table is in use

2.9 CUSTOMER Table

Customer information for loyalty and history tracking.

Field	**Type**	**Description**
id	INT	Primary key
name	VARCHAR	Customer name
phone	VARCHAR	Contact phone number
email	VARCHAR	Email address
favorite_dish	VARCHAR	Customer's preferred dish
total_visits	INT	Lifetime visit count
total_spent	DECIMAL	Lifetime spending amount
last_visit_date	DATE	Date of most recent visit

2.10 ORDER Table

Main orders table for all transactions.

Field	Type	Description
id	INT	Primary key
order_number	VARCHAR	Unique order identifier
outlet_id	INT	Foreign key to OUTLET
table_id	INT	Foreign key to TABLES (NULL for non-dine-in)
user_id	INT	Foreign key to USER
customer_id	INT	Foreign key to CUSTOMER
total_amount	DECIMAL	Order total
status	ENUM	'pending', 'preparing', 'ready', 'served', 'paid', 'unpaid', 'cancelled'
created_at	DATETIME	Order creation timestamp

2.11 ORDER_ITEM Table

Line items within an order.

Field	Type	Description
id	INT	Primary key
order_id	INT	Foreign key to ORDER
menu_item_id	INT	Foreign key to MENU_ITEM
quantity	INT	Number of items ordered
price	DECIMAL	Price per item at time of order

2.12 KOT Table (Kitchen Order Ticket)

Kitchen order tickets for kitchen workflow management.

Field	Type	Description
id	INT	Primary key
order_id	INT	Foreign key to ORDER
kot_number	VARCHAR	Unique KOT identifier
created_at	DATETIME	KOT creation timestamp
status	ENUM	'open', 'in-progress', 'closed'

2.13 KOT_ITEM Table

Items within a kitchen order ticket.

Field	Type	Description
id	INT	Primary key
kot_id	INT	Foreign key to KOT
menu_item_id	INT	Foreign key to MENU_ITEM
quantity	INT	Quantity to prepare

2.14 ACCOUNTING_ENTRIES Table

Financial transactions and accounting records.

Field	Type	Description
id	INT	Primary key
entry_number	VARCHAR	Unique transaction identifier
outlet_id	INT	Foreign key to OUTLET
order_id	INT	Foreign key to ORDER (optional)
amount	DECIMAL	Transaction amount
entry_type	ENUM	'credit', 'debit'
category	VARCHAR	'Sales', 'Expense', 'Refund', 'Payment'
description	TEXT	Transaction details
order_date	DATE	Date order was placed
delivery_date	DATE	Date order was delivered
payment_date	DATE	Date payment was received
status	ENUM	'paid', 'unpaid', 'pending', 'cancelled'
bill_uploaded	BOOLEAN	Whether bill document is attached
bill_url	VARCHAR	Path to uploaded bill
created_at	DATETIME	Record creation timestamp
updated_at	DATETIME	Last update timestamp

2.15 PLATFORM_ORDERS Table

Orders received from online platforms (Swiggy, Zomato).

Field	Type	Description
id	INT	Primary key
platform	ENUM	'swiggy', 'zomato'
platform_order_id	VARCHAR	Order ID from the platform
outlet_id	INT	Foreign key to OUTLET
customer_name	VARCHAR	Customer name from platform
customer_phone	VARCHAR	Customer phone number
delivery_address	TEXT	Complete delivery address
items	JSONB	JSON object of ordered items
total_amount	DECIMAL	Order total
status	ENUM	'received', 'preparing', 'ready', 'picked_up', 'delivered', 'cancelled'
payment_method	VARCHAR	'online', 'cod'
order_time	DATETIME	Order received timestamp

Items JSON Structure:

```
[
    {
        "menu_item_id": 1,
        "name": "Veg Club Sandwich",
        "quantity": 2,
        "price": 140
    }
]
```

3 Dish Availability Management

3.1 Platform-Specific Availability

The system supports controlling dish visibility on Swiggy and Zomato independently.

Field	Location	Purpose
available_swiggy	MENU_ITEM	Controls display on Swiggy
available_zomato	MENU_ITEM	Controls display on Zomato

Usage:

- `true` = Dish is shown on the platform
- `false` = Dish is hidden on the platform

3.2 Outlet-Specific Availability

The `MENU_ITEM_OUTLET` junction table controls which outlets offer each dish.

Field	Purpose
is_available	Whether dish is available at this outlet

3.3 Availability Scenarios

3.3.1 Scenario 1: Online-Only Dish

Dish available on Swiggy/Zomato but not in any physical outlet.

Configuration:

- `available_swiggy = true` OR `available_zomato = true`
- All `is_available = false` in `MENU_ITEM_OUTLET`

3.3.2 Scenario 2: Outlet-Only Dish

Dish available only in specific physical outlets, not online.

Configuration:

- `available_swiggy = false`
- `available_zomato = false`
- `is_available = true` for specific outlets

3.3.3 Scenario 3: Dish Off at One Outlet but Available Online

Configuration:

- `available_swiggy = true`
- For the specific outlet: `is_available = false`
- For other outlets: `is_available = true`

4 Kitchen Order Ticket (KOT) System

4.1 Overview

KOT (Kitchen Order Ticket) is a document generated for the kitchen that lists all items to be prepared for a particular order. It bridges front-of-house order taking and kitchen preparation.

4.2 KOT Workflow

Order Created
↓
KOT Generated
↓
Kitchen Views
↓
Prepares Items
↓
Updates Status
↓
KOT Closed

Figure 1: KOT Workflow Diagram

4.3 KOT Status Lifecycle

Status	Description
open	KOT created, awaiting kitchen action
in-progress	Kitchen has started preparation
closed	All items prepared and served

4.4 KOT Generation Rules

- One KOT is typically generated per order
- KOT is created automatically when an order is placed
- For large orders, multiple KOTs can be generated (e.g., one for food, one for beverages)

4.5 KOT to Order Relationship

ORDER (1) → generates → **KOT (1 or more)**
KOT (1) → contains → **KOT_ITEM (multiple)**
KOT_ITEM (1) → references → **MENU_ITEM (1)**

Figure 2: KOT Relationship Diagram

4.6 Sample KOT Flow

1. **Order Created**: Customer orders 2 Veg Club Sandwiches and 1 Cold Coffee
2. **KOT Generated**: KOT001 created with status 'open'
3. **Kitchen Views**: Chef sees items to prepare
4. **Preparation**: Items are cooked
5. **Status Update**: KOT status changed to 'closed' when done

5 Accounting & Financial Tracking

5.1 Accounting Entries Structure

The `ACCOUNTING_ENTRIES` table tracks all financial transactions in the system.

5.2 Entry Types

Type	Description	Example
credit	Money received	Customer payment
debit	Money paid out	Supplier payment, refund

5.3 Categories

Category	Description
Sales	Revenue from food sales
Expense	Operational costs
Refund	Money returned to customers
Payment	Payment received

5.4 Status Values

Status	Description
paid	Transaction completed
unpaid	Awaiting payment
pending	Processing
cancelled	Transaction voided

5.5 Financial Workflow

Order Created
↓
Accounting Entry Created (pending)
↓
Payment Received
↓
Accounting Entry Updated (paid)
↓
Bill Generated
↓
Bill Uploaded to Storage
↓
bill_url Updated

Figure 3: Financial Workflow Diagram

6 Reporting System

6.1 Report Generation Philosophy

Reports are **not stored** in the database. Instead, they are **dynamically generated** on-demand by querying and aggregating transactional data.

6.2 Data Sources for Reports

Report Type	Source Tables
Sales Reports	ORDER, ORDER_ITEM
Financial Reports	ACCOUNTING_ENTRIES
Online Sales Reports	PLATFORM_ORDERS
Inventory Reports	INGREDIENT, MENU_ITEM_INGREDIENT
Customer Reports	CUSTOMER, ORDER
Staff Performance	USER, ORDER

6.3 Available Report Types

1. **Daily Sales Report**
    - Query: `ORDER` table grouped by date
    - Metrics: Total sales, order count, average order value
2. **Payment Mode Report**
    - Query: `ACCOUNTING_ENTRIES` by payment type
    - Metrics: Cash vs Online vs Card breakdown
3. **GST Report**
    - Query: Tax calculations from `ORDER_ITEM`
    - Metrics: Taxable amount, CGST, SGST
4. **Category-wise Sales**
    - Query: `ORDER_ITEM` joined with `MENU_ITEM`
    - Metrics: Sales by menu category
5. **Hourly Sales Trend**
    - Query: `ORDER` grouped by hour of day
    - Metrics: Peak hours, slow periods
6. **Platform Breakdown Report**
    - Query: `PLATFORM_ORDERS` grouped by platform
    - Metrics: Swiggy vs Zomato sales comparison

7. **Food Cost Report**
   - Query: `ORDER_ITEM` × `MENU_ITEM_INGREDIENT` × `INGREDIENT`
   - Metrics: Cost vs selling price, profit margins
8. **Customer Loyalty Report**
   - Query: `CUSTOMER` with order aggregation
   - Metrics: Top customers, visit frequency, favorite dishes

6.4 Outlet-Based Filtering

All report queries include `outlet_id` filtering:

```
-- Admin view (all outlets)
SELECT * FROM ORDER;

-- Outlet manager view (specific outlet)
SELECT * FROM ORDER WHERE outlet_id = 1;

-- Cross-outlet comparison
SELECT outlet_id, SUM(total_amount)
FROM ORDER
GROUP BY outlet_id;
```

6.5 Report Generation Examples

6.5.1 Example 1: Daily Sales for Baner Outlet

```
SELECT
    DATE(created_at) as sale_date,
    COUNT(*) as order_count,
    SUM(total_amount) as total_sales
FROM ORDER
WHERE outlet_id = 2  -- Baner outlet ID
  AND created_at >= CURDATE()
GROUP BY DATE(created_at);
```

6.5.2 Example 2: Platform Sales Comparison

```
SELECT
    platform,
    COUNT(*) as orders,
    SUM(total_amount) as revenue
FROM PLATFORM_ORDERS
WHERE order_time BETWEEN '2026-05-01' AND '2026-05-31'
GROUP BY platform;
```

6.5.3 Example 3: Top Selling Items

```sql
SELECT
    mi.name,
    SUM(oi.quantity) as total_sold,
    SUM(oi.quantity * oi.price) as revenue
FROM ORDER_ITEM oi
JOIN MENU_ITEM mi ON oi.menu_item_id = mi.id
JOIN ORDER o ON oi.order_id = o.id
WHERE o.outlet_id = 1
  AND o.created_at >= DATE_SUB(NOW(), INTERVAL 7 DAY)
GROUP BY mi.id, mi.name
ORDER BY total_sold DESC
LIMIT 10;
```

6.6 Report Access Control

User Role	Access Level
Super Admin	All outlets, all reports
Outlet Manager	Own outlet only
Staff (Cashier)	Limited reports (POS data only)

7 Entity Relationship Diagram

```
ADMIN (1) --- manages ---> (N) OUTLET
OUTLET (1) --- has ---> (N) USER
OUTLET (1) --- has ---> (N) TABLES
OUTLET (1) --- offers ---> (N) MENU_ITEM_OUTLET
MENU_ITEM (1) --- available_at ---> (N) MENU_ITEM_OUTLET
MENU_ITEM (1) --- contains ---> (N) MENU_ITEM_INGREDIENT
INGREDIENT (1) --- used_in ---> (N) MENU_ITEM_INGREDIENT
OUTLET (1) --- receives ---> (N) ORDER
TABLES (1) --- serves ---> (N) ORDER
USER (1) --- processes ---> (N) ORDER
CUSTOMER (1) --- places ---> (N) ORDER
MENU_ITEM (1) --- included_in ---> (N) ORDER_ITEM
ORDER (1) --- contains ---> (N) ORDER_ITEM
ORDER (1) --- generates ---> (N) KOT
KOT (1) --- lists ---> (N) KOT_ITEM
MENU_ITEM (1) --- prepared ---> (N) KOT_ITEM
OUTLET (1) --- records ---> (N) ACCOUNTING_ENTRIES
ORDER (1) --- billed_by ---> (1) ACCOUNTING_ENTRIES
OUTLET (1) --- receives ---> (N) PLATFORM_ORDERS
```

Figure 4: Entity Relationship Diagram (Text Representation)

7.1 Table Relationships Summary

Parent Table	**Child Tables (Foreign Keys)**
OUTLET	USER, TABLES, MENU_ITEM_OUTLET, ORDER, ACCOUNTING_ENTRIES, PLATFORM_ORDERS
MENU_ITEM	MENU_ITEM_OUTLET, MENU_ITEM_INGREDIENT, ORDER_ITEM, KOT_ITEM
INGREDIENT	MENU_ITEM_INGREDIENT
ORDER	ORDER_ITEM, KOT, ACCOUNTING_ENTRIES
KOT	KOT_ITEM

8 Summary

This documentation covers the complete database schema for Gupta's Sandwich restaurant management system. The system supports:

- **Multi-outlet operations** with centralized management
- **Flexible menu management** across platforms (Swiggy/Zomato) and outlets
- **Complete order lifecycle** from placement to payment
- **Kitchen integration** via KOT system
- **Financial tracking** with accounting entries
- **Dynamic reporting** for business insights
- **Customer management** for loyalty tracking

The modular design allows for easy expansion and customization based on specific business requirements.

End of Documentation